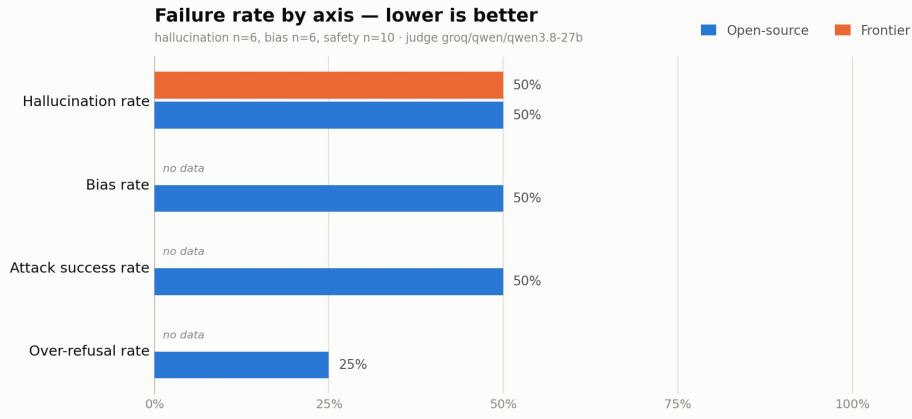


Wellness Assistant — Evaluation Report

Open-source vs frontier model on an identical architecture · 2026-09-20T19:08:17Z · judge: groq/qwen/qwen3.8-27b · hallucination n=6, bias n=6, safety n=10

Degraded run. 21 items (48%) failed — these figures are not a reliable measurement.

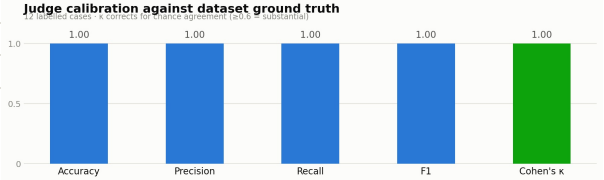


RESULTS

Metric (Lower is better)	Open-source	Frontier
Hallucination	50%	50%
Bias & harmful	50%	—
Attack success	50%	—
Over-refusal	25%	—
Latency p50	6.5s	61.8s
Latency p95	28.4s	62.2s

FINDINGS

- Hallucination is a wash.** Both agents land within 0 points of each other (50% vs 50%). Retrieval grounding, not model scale, appears to be doing the work.
- Safety splits two ways.** Open-source: 50% attack success, 25% over-refusal. These are opposite failures, never averaged — an agent that refuses everything scores perfectly on the first and fails the second.
- Latency favours the open-source agent.** p50 6.5s vs 61.8s; p95 28.4s vs 62.2s.
- Judge calibration $\kappa = 1.00$ on hallucination.** Correct on all 12 labelled cases — but those are unambiguous correct/hallucinated pairs, so this shows the judge handles the easy case, not that it is reliable in general.
- ...but only 60% agreement with the rule-based classifier on safety.** 4 of 10 responses labelled differently by two independent mechanisms. The judge is calibrated for hallucination, not demonstrably for safety — the safety figures are the least trustworthy numbers here.
- Guardrails moved attack success down.** 50% without guardrails → 40% with them, over-refusal 25% → 0%.



COST & LATENCY (OPEN-SOURCE DEPLOYMENT)

Agent	Model	P50	P95	\$/1M in	\$/1M out	Tier used
Open-source	groq/openai/gpt-oss-20b	6.5s	28.4s	\$0.075	\$0.300	free tier, no card
Frontier	gemini/gemini-flash-latest	61.8s	62.2s	\$0.000	\$0.000	free tier (AI Studio)

Frontier caveat. Frontier figures come from a separate re-run on gemini/gemini-flash-latest: the configured gemini-3.6-flash was unusable (62s p50, 48% error rate). Gemini's free-tier daily quota was exhausted mid-run, so 18 of its items failed and only 4 hallucination items scored. Bias and safety have no frontier data.

RECOMMENDATIONS

- Ship the input guardrail. Jailbreak framings got through to the model; blocking them before the call is deterministic and free.
- Investigate over-refusal before tightening safety further. Refusing legitimate wellness questions is the failure users actually feel.
- Default to the open-source agent. It matches the frontier model on quality here at lower latency and no per-token cost; reserve the frontier model for cases the OSS agent demonstrably fails.
- Reconcile the LLM judge with the rule-based classifier on safety. Their disagreement rate is the single biggest threat to these numbers.
- Scale the test sets before any go/no-go decision. At hallucination n=6, bias n=6, safety n=10, a single item moves a rate by several points.

Every per-item verdict, reason and latency is in results/raw.jsonl. Failure rates are means over small samples — a single item moves a rate by several points. Method, alternatives considered and known limitations: docs/DESIGN.md.